Abdelrahman Elewah

Oshawa, Ontario, Canada | abdelrahman.elewah@gmail.com | +1 289-939-6665 | Portfolio: elewah.github.io

LinkedIn: linkedin.com/in/abdelrahman-elewah | GitHub: github.com/elewah | Google Scholar

Summary

AI Integration Lead and AI/ML Engineer specializing in production AI systems, LLM/RAG applications, computer vision, IoT platforms, and scalable data pipelines. Strong background translating applied research into reliable software solutions using Python, LangGraph, FastAPI, PostgreSQL, Docker, AWS, and modern machine learning frameworks.

Technical Skills

AI/ML and LLMs: LLMs, RAG, Agentic AI, LangChain, LangGraph, LangSmith, Docling, MinerU, MCP, semantic search, computer vision, document intelligence, model deployment
Programming Languages: Python, C/C++, JavaScript
Web and APIs: FastAPI, Flask, REST APIs, React, HTML, CSS, Bootstrap
Cloud, MLOps, and DevOps: AWS EC2, AWS App Runner, Amazon ECS, Docker, Docker Compose, Kubernetes, CI/CD, Jenkins, GitLab CI/CD, GitHub Actions
Data Systems: PostgreSQL, MongoDB, MySQL, NoSQL, SQL, Spark, Hadoop
Development Tools: GitHub, GitLab, VS Code, Anaconda, Dev Containers

Professional Experience

Artificial Intelligence Integration Lead, MPAC — Pickering, ON — Jul 2025 – Present

- Lead enterprise AI and automation integration across property assessment workflows, improving digital operations through production-ready AI/ML systems.
- Architect scalable data pipelines with Python, LangGraph, FastAPI, Docling, MinerU, PostgreSQL, and MCP servers for building plan parsing, document intelligence, and semantic search.
- Design and deploy RAG systems for building plan understanding, retrieval workflows, and assessment-standard knowledge access.
- Integrate vision-language models such as Qwen3-VL, LayoutLMv3, PaddleOCR-VL, and PP-StructureV3 with RF-DETR for computer vision, object detection, and layout understanding.

Instructional Specialist (part-time), 2U / University of Toronto — Remote — Jan 2023 – Apr 2025

- Contributed to the success of the **University of Toronto**'s online **Data Analytics Boot Camp**, supporting 100+ learners in mastering practical skills for data-driven careers.
- Facilitated hands-on workshops in **Python, Database, Machine Learning, and Data Visualization**, resulting in a **15% increase** in student satisfaction scores.

Research Assistant, Ontario Tech University — Oshawa, ON — Jan 2020 – Apr 2025

- Designed the **SensorsConnect** framework, enabling real-time **IoT device interoperability** and scalable data exchange, inspired by the principles of the **World Wide Web**. Publication
- Developed an **Agentic IoT Search Engine (ASE-IoT)** that integrates **LLMs**, **RAG**, and autonomous agents to enable **natural language querying** of live IoT data, enhancing search precision and user interaction. Publication
- Co-led an **OVIN-funded project** on **software-defined vehicles (SDVs)** and **digital twins**, contributing to workforce development in emerging mobility technologies.

Back-End Developer (part-time), Tamra-IoT — Toronto, ON — May 2019 – Jan 2023

- **Architected** secure and scalable **IoT platforms** by integrating **MQTT over TLS**, **cloud infrastructure**, and **mobile control interfaces**, enhancing real-time communication and remote device management.
- **Designed** and deployed **Over-The-Air (OTA) firmware update mechanisms** and implemented **robust IoT device management systems**, significantly reducing maintenance costs and improving system resilience. Publication

Education

PhD Ontario Tech University, Electrical and Computer Engineering Jan 2020 – Mar 2025

- GPA: 4.22/4.3 Link to Transcript issued by Ontario Tech University
- **Thesis**: SensorsConnect: World Wide Web for Internet of Things.

MSc Benha University, Electrical Engineering Feb 2013 – Jan 2018
BSc Benha University, Electrical Engineering Sep 2008 – Jun 2012

Projects

Localelive: Agentic Search Engine for Real-Time IoT Data | GitHub | Live Demo | Publication

- **Developed** a real-time IoT search engine powered by **LLMs and RAG**, enabling users to query complex sensor data using natural language, improving query efficiency and decision making.
- **Implemented** a semantic search pipeline using **Sentence-BERT and HNSW indexing**, reducing query latency by 73% and enhancing relevance in top-k retrieval across diverse IoT datasets.
- **Managed** over 37,000 real-time IoT documents from 500+ service types in **MongoDB** with geo-indexing, ensuring scalable and location-aware data access for time-sensitive decision-making.
- **Achieved** 92% top-1 accuracy in complex intent detection and information retrieval, surpassing systems like Gemini, and significantly improving user satisfaction and task completion rates.
- **Deployed** on **AWS** using FastAPI, LangGraph, Docker Compose, and Traefik for modular retrieval and production-grade orchestration.

Story-to-Movie Recommender Chatbot (RAG-based) | GitHub | Live Demo

- Developed a **retrieval-augmented generation (RAG)** system combining vector search with LLMs to deliver context-aware answers, reducing hallucination rates by **30%**.
- Built a **semantic search pipeline**, enabling retrievals from 1K+ documents.
- Leveraged **Pandas**, **OpenAI API**, and **Tenacity** to ensure resilient API usage and robust data handling under real-time loads.

Apply Lightweight Fine-Tuning to a Foundation Model | GitHub

- **Built** an end-to-end NLP pipeline using **PyTorch**, **TensorFlow** and **Hugging Face Transformers**: loaded a pre-trained **GPT-2** model and prepared the **AG News** dataset for news classification.
- **Applied** LoRA-based parameter-efficient fine-tuning (PEFT), reducing training time and memory usage by over 60% while improving AG News accuracy from 83.16% to 88.95%.

RadViz-Plotly | GitHub

- **Developed RadViz-Plotly**, an **open-source Python package** that generates **2D and 3D Radial Visualization (RadViz)** plots for **high-dimensional datasets**, enabling broader accessibility to dimensionality reduction techniques in research and industry. Publication
- **Enabled** data scientists to explore complex distributions interactively using **Plotly**, improving model explainability and analytics workflows.